

Math 170E
Week 5

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1 Continuous Random Variables & Poisson Distribution

Example 1. On average, how many rolls we need to throw a fair dice to get all 6 outcomes?

- (A)6 (B)36 (C)12 (D)14.7 (E) $\frac{144}{7}$

Example 2. (PSI 2.7.5)

1. Flaws in a certain type of drapery material appear on the average of one in 150 square feet. If we assume a Poisson distribution, approximate the probability of at most one flaw appearing in 225 square feet.

- (A)0.06 (B)0.12 (C)0.33 (D)0.56 (E)0.67

2. **(modified version)** In the same material, what is the probability that we see at most two flaws in 300 square feet?

- (A) e^{-2} (B) $5e^{-1}$ (C) $5e^{-2}$ (D) $6e^{-3}$ (E) $10e^{-\frac{3}{2}}$

Example 3. (PSI 3.1.15) The life X (in years) of a voltage regulator of a car has the pdf

$$f(x) = \frac{3x^2}{7^3} e^{-(x/7)^3}$$

for $0 < x < \infty$.

1. What is the probability that this regulator will last at least 7 years?

(A) e^{-1} (B) e^{-7} (C) $e^{-1/7}$ (D) $\frac{1}{7}$ (E) $\frac{316}{343}$

2. Given that it has lasted at least 7 years, what is the conditional probability that it will last at least another 3.5 years?

(A) e^{-1} (B) $e^{-19/8}$ (C) $e^{-27/8}$ (D) $\frac{7}{8}$ (E) $\frac{27}{343}$

Example 4. (Old MT problem) Let us consider an unfair coin which tosses head and tail with probability 0.01 and 0.99, respectively.

1. Toss this coin independently, and let X be the first time that head appears. Compute the mean of X .

(A) 1 (B) $\frac{1}{100}$ (C) $\frac{1}{99}$ (D) 99 (E) 100

2. Compute the variance of X .

(A) 99 (B) 100 (C) 9900 (D) $\frac{10000}{99}$ (E) 10000

3. Toss this coin for 500 times independently. Use a Poisson limit theorem to compute (approximately) the probability that exactly 5 many head appears.

(A) 99 (B) 100 (C) 9900 (D) $\frac{10000}{99}$ (E) 10000

Example 5. (Old Quiz Problem) Let X be a continuous random variable with uniform distribution $U([3, 4])$.

1. Compute $\mu = EX$

(A) 3 (B) 3.5 (C) 4 (D) 7 (E) 49

2. Compute $M(t)$

(A) t^7 (B) $e^{7t/2}$ (C) $\frac{e^{7t/2}}{12}$ (D) $\frac{e^{4t} - e^{3t}}{12}$ (E) $\frac{e^{4t} - e^{3t}}{t}$

3. Compute $\sigma^2 = \text{Var}(X)$

- (A) 1 (B) 12 (C) 7 (D) $\frac{1}{7}$ (E) $\frac{1}{12}$

Example 6. (Old MT problem) Suppose that a coin is not fair so that the probability of obtaining a head is $p \in (0, 1)$.

1. On average, how many flips are needed to obtain a head?
2. Find the probability the first time obtaining a head is an even number. Your final answer must not be an infinite series.